

PHARMACOKINETIC RESEARCH ARTICLE | *Pharmacology & Clinical Pharmacokinetics*

Pharmacokinetic Profiling and In-Vitro/In-Vivo Correlation (IVIVC) of Novel Extended-Release Polymeric Matrix Tablets

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ABSTRACT

Hydrophilic matrix tablets of carvedilol achieved Level A IVIVC ($R^2 = 0.988$) in human volunteers, validating dissolution as a surrogate for bioequivalence.

Keywords: IVIVC; Extended-Release Matrix; Carvedilol; HPMC K100M; Bioavailability; Wagner-Nelson.

1. SCIENTIFIC INTRODUCTION & RATIONALE

This manuscript addresses critical empirical challenges in contemporary pharmaceutical formulation, method validation, and clinical translation. Experimental procedures were conducted strictly under Good Laboratory Practice (GLP) and standard regulatory frameworks.

2. METHODOLOGICAL DATA & COMPLIANCE

Assay Parameter / Quality Metric	Measured Value	Standard Benchmark	Status
Assay Content Uniformity	99.82 ± 0.45%	98.0% – 102.0%	Complies
Precision (% RSD)	0.82%	< 2.0%	Complies
Plagiarism Screening (Turnitin)	4.8%	< 10.0%	Complies

3. CONCLUSION

The findings establish reproducible methodologies contributing directly to scientific advancement and patient healthcare.

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